

Proof That the First Variation of Entropy Is $\log \rho + 1$

Statement

Let

$$\mathcal{H}(\rho) = \int_{\mathbb{R}^d} \rho(x) \log \rho(x) dx$$

be the Boltzmann entropy functional, using the convention common in Wasserstein gradient flows. Assume ρ is positive and smooth enough for the following computation, and let η be a smooth perturbation satisfying

$$\int_{\mathbb{R}^d} \eta(x) dx = 0.$$

Then

$$\frac{\delta \mathcal{H}}{\delta \rho} = \log \rho + 1.$$

Proof

Consider the perturbed density

$$\rho_\varepsilon = \rho + \varepsilon \eta.$$

For sufficiently small ε , this remains positive if ρ is strictly positive and η is bounded. We compute

$$\left. \frac{d}{d\varepsilon} \mathcal{H}(\rho_\varepsilon) \right|_{\varepsilon=0}.$$

By definition,

$$\mathcal{H}(\rho_\varepsilon) = \int_{\mathbb{R}^d} (\rho(x) + \varepsilon \eta(x)) \log(\rho(x) + \varepsilon \eta(x)) dx.$$

Differentiate the scalar function

$$h(r) = r \log r.$$

Its derivative is

$$h'(r) = \log r + 1.$$

Therefore, pointwise in x ,

$$\left. \frac{d}{d\varepsilon} \left[(\rho(x) + \varepsilon \eta(x)) \log(\rho(x) + \varepsilon \eta(x)) \right] \right|_{\varepsilon=0} = (\log \rho(x) + 1) \eta(x).$$

Integrating gives

$$\left. \frac{d}{d\varepsilon} \mathcal{H}(\rho + \varepsilon \eta) \right|_{\varepsilon=0} = \int_{\mathbb{R}^d} (\log \rho(x) + 1) \eta(x) dx.$$

By the definition of first variation, this means

$$\frac{\delta \mathcal{H}}{\delta \rho}(\rho)(x) = \log \rho(x) + 1.$$

Remark on the Constant

When working on probability densities, admissible perturbations satisfy

$$\int_{\mathbb{R}^d} \eta(x) \, dx = 0.$$

Hence adding any constant C to the first variation does not change the directional derivative, because

$$\int_{\mathbb{R}^d} C \eta(x) \, dx = 0.$$

Thus, on the probability simplex, the first variation is defined only up to an additive constant. One often writes simply

$$\frac{\delta \mathcal{H}}{\delta \rho} = \log \rho$$

when the constant is irrelevant. In Wasserstein gradient flows, the constant also disappears after taking a spatial gradient:

$$\nabla(\log \rho + 1) = \nabla \log \rho.$$